

Tutorial 4: Undecidability, Reducibility, Rice's Theorems

Problem 1 For a language L over the alphabet $\{0, 1\}$, define the language

$$\text{HALF}(L) = \{x \mid x \in \Sigma^*, \text{ and there exists } y \in \Sigma^* \text{ such that } |x| = |y| \text{ and } xy \in L\}$$

Prove/Disprove the following statements.

- (a) If L is R.E., then $\text{HALF}(L)$ must be R.E.
 - (b) If L is recursive, then $\text{HALF}(L)$ must be recursive.
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Solution (a) Let M be a DTM recognising L .

Create an NTM N for $\text{HALF}(L)$:

N on input x :

- Nondeterministically choose $y \in \Sigma^*$ such that $|x| = |y|$.
- Simulate M on xy .
- If M accepts, accept x .

Thus, $\text{HALF}(L)$ is R.E.

(b) Let M be a total TM deciding L .

Create a total TM N for $\text{HALF}(L)$:

N on input x :

- Check for all $y \in \Sigma^*$ and $|x| = |y|$:
 - Simulate M on xy .
- If M accepts in any of the cases, accept x . Otherwise, reject.

Thus, $\text{HALF}(L)$ is recursive. □

Problem 2 Prove that the following language is not recursive:

$$\text{WB} = \{M\#w \mid M \text{ writes the blank symbol in some step on input } w\}$$

Solution We show a reduction $\text{HP} \leq_m \text{WB}$:

$$N\#v \mapsto M\#w$$

$$N \text{ halts on } v \iff M \text{ writes the blank symbol in some step on } w$$

M on input w :

- Simulate N on v .
 - Use a different symbol if N tries to write the blank symbol.
- If N halts, write the (actual) blank symbol on the input.

Thus, WB is not recursive. □

Problem 3 Prove:

- (a) $E_{2026} = \{M \mid M \text{ halts on exactly 2026 inputs}\}$ is not R.E.
 - (b) $\text{AL}_{2026} = \{M \mid M \text{ halts on at least 2026 inputs}\}$ is R.E. but not recursive.
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Solution (a) Show a reduction $\overline{\text{HP}} \leq_m E_{2026}$:

$$M\#w \mapsto N$$

$$M \text{ does not halt on } w \iff N \text{ halts on exactly 2026 inputs}$$

N on input v :

- If $v = 0^k$, $k < 2026$, accept.
- Simulate M on w .
- If M halts, accept.

Denote by $L_H(N)$, the set of all inputs on which N halts. Then,

$$L_H(N) = \begin{cases} \Sigma^* & \text{if } M \text{ halts on } w \\ \{0^k \mid k < 2026\} & \text{if } M \text{ does not halt} \end{cases}$$

which satisfies the reduction constraint.

Thus, E_{2026} is not R.E.

(b) Create a TM N recognising AL_{2026} :

N on input M :

- Simulate M on all inputs on a time sharing basis.
- If M halts on at least 2026 inputs, accept.

Thus, AL_{2026} is R.E.

Show a reduction from $HP \leq_m AL_{2026}$:

$$M\#w \mapsto N$$

$$M \text{ halts on } w \iff N \text{ halts on at least 2026 inputs}$$

N on input v :

- Simulate M on w .
- If M halts, accept v .

Here,

$$L_H(N) = \begin{cases} \Sigma^* & \text{if } M \text{ halts on } w \\ \phi & \text{otherwise} \end{cases}$$

Thus, the reduction is correct, and AL_{2026} is not recursive. □

Problem 4 Let k be a constant positive integer. Consider the following languages:

- $LE_k = \{M \mid \text{DTM } M \text{ loops on at most } k \text{ inputs}\}$
- $LT_k = \{M \mid \text{DTM } M \text{ loops on less than } k \text{ inputs}\}$
- $GE_k = \{M \mid \text{DTM } M \text{ loops on at least } k \text{ inputs}\}$
- $GT_k = \{M \mid \text{DTM } M \text{ loops on more than } k \text{ inputs}\}$

Prove that all these languages are non-R.E. and non-co-R.E.

Solution First note that $\overline{LE_k} = GT_k$, $\overline{LT_k} = GE_k$. So we only need to show:

- $\overline{HP} \leq_m LE_k$ and $\overline{HP} \leq_m GT_k$ (LE_k and GT_k are non-R.E. and non-co-R.E.)
- $\overline{HP} \leq_m LT_k$ and $\overline{HP} \leq_m GE_k$ (LT_k and GE_k are non-R.E. and non-co-R.E.)

$\overline{HP} \leq_m LE_k$:

$$M\#w \mapsto N$$

$$M \text{ does not halt on } w \iff N \text{ loops on at most } k \text{ inputs}$$

$$M \text{ does not halt on } w \Rightarrow N \text{ loops on at most } k \text{ inputs}$$

$$M \text{ halts on } w \Rightarrow N \text{ loops on more than } k \text{ inputs}$$

N on input v :

- Simulate M on w for $|v|$ steps.
- If M did not halt yet, accept (and halt). Otherwise, loop.

Let K' denote the set of strings on which N loops.

If M does not halt on w , then $K' = \phi$. If M halts on w in s steps, then, $K' = \{v \mid |v| \geq s\}$. So,

$$|K'| = \begin{cases} 0 & \text{if } M \text{ does not halt on } w \\ \infty & \text{if } M \text{ halts on } w \end{cases}$$

which satisfies our reduction constraint. Hence, $\overline{\text{HP}} \leq_m \text{LE}_k$.

(The same construction also shows $\overline{\text{HP}} \leq_m \text{LT}_k$.)

$\overline{\text{HP}} \leq_m \text{GT}_k$:

$$M \# w \mapsto N$$

M does not halt on $w \iff N$ loops on more than k inputs

M does not halt on $w \Rightarrow N$ loops on more than k inputs

M halts on $w \Rightarrow N$ loops on at most k inputs

N on input v :

- Simulate M on w .
- If M halts on w , accept v .

If M halts on w , $K' = \phi$. If M does not halt, then $K' = \Sigma^*$. So

$$|K'| = \begin{cases} \infty & \text{if } M \text{ does not halt on } w \\ 0 & \text{if } M \text{ halts on } w \end{cases}$$

which satisfies our reduction constraint. Hence, $\overline{\text{HP}} \leq_m \text{GT}_k$.

(The same construction also shows $\overline{\text{HP}} \leq_m \text{GE}_k$.) □

Problem 5 Let $\text{nsteps}(M, w)$ denote the number of steps of M on w . If M loops on w , take $\text{nsteps}(M, w) = \infty$. If N also loops on v , take $\text{nsteps}(M, w) = \text{nsteps}(N, v)$.

Prove: recursive / R.E but not recursive / non-R.E.?

- (a) $L_a = \{M \# N \mid \text{nsteps}(M, \varepsilon) < \text{nsteps}(N, \varepsilon)\}$
 - (b) $L_b = \{M \# N \mid \text{nsteps}(M, \varepsilon) \leq \text{nsteps}(N, \varepsilon)\}$
 - (c) $L_c = \{M \# N \mid \text{nsteps}(M, w) < \text{nsteps}(N, v) \text{ for some } w, v\}$
 - (d) $L_d = \{M \# N \mid \text{nsteps}(M, w) < \text{nsteps}(N, v) \text{ for all } w, v\}$
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Solution (a) R.E. but not recursive.

To create a recogniser for L_a , simulate both M and N on ε parallelly, and if M halts earlier, then accept.

To show $\text{HP} \leq_m L_a$:

$$K \# w \mapsto M \# N$$

K halts on $w \iff \text{nsteps}(M, \varepsilon) < \text{nsteps}(N, \varepsilon)$

$$K \text{ halts on } w \Rightarrow \text{nsteps}(M, \varepsilon) < \text{nsteps}(N, \varepsilon)$$

$$K \text{ does not halt on } w \Rightarrow \text{nsteps}(M, \varepsilon) \geq \text{nsteps}(N, \varepsilon)$$

M on input v_1 :

- Simulate K on w .
- If it halts, accept.

N on input v_2 :

- Loop unconditionally.

Here, we have

- If K halts on w , then $\text{nsteps}(M, \varepsilon) = \text{finite}$, $\text{nsteps}(N, \varepsilon) = \infty$; so $\text{nsteps}(M, \varepsilon) < \text{nsteps}(N, \varepsilon)$.
- If K does not halt on w , then $\text{nsteps}(M, \varepsilon) = \infty = \text{nsteps}(N, \varepsilon)$; so $\text{nsteps}(M, \varepsilon) \geq \text{nsteps}(N, \varepsilon)$.

So, L_a is non recursive.

(b) non-R.E.

Intuitively, we cannot create a recogniser for L_b , because in the case where $\text{nsteps}(M, \varepsilon) = \text{nsteps}(N, \varepsilon) = \infty$, we cannot accept $M\#N$ in finite time. (Their simulation will keep on looping and never halt, so our recogniser can't accept the input.)

To show $\overline{\text{HP}} \leq_m L_b \equiv \text{HP} \leq_m \overline{L_b}$.

Note that $\overline{L_b} = \{M\#N \mid \text{nsteps}(M, \varepsilon) > \text{nsteps}(N, \varepsilon)\}$ is essentially the same as L_a , just with the roles of M and N reversed. So the same construction as in (a) can be used (by switching M and N) to show $\text{HP} \leq_m \overline{L_b}$.

Thus, L_b is non-R.E.

(c) R.E. but non-recursive.

Consider an NTM which nondeterministically chooses some w and v to run M and N on (parallelly), and if M halts before N , then accepts. This is a recogniser for L_c .

$\text{HP} \leq_m L_c$ follows from the same construction as (a).

$$K\#x \mapsto M\#N$$

$$K \text{ halts on } x \iff \exists w, v \quad \text{nsteps}(M, w) < \text{nsteps}(N, v)$$

$$K \text{ halts on } x \Rightarrow \exists w, v \quad \text{nsteps}(M, w) < \text{nsteps}(N, v)$$

$$K \text{ does not halt on } x \Rightarrow \forall w, v \quad \text{nsteps}(M, w) \geq \text{nsteps}(N, v)$$

From the construction of M and N in (a),

- If K halts on x , then $\text{nsteps}(M, w) = \text{finite}$, $\text{nsteps}(N, v) = \infty$; so $\text{nsteps}(M, w) < \text{nsteps}(N, v)$, for any w, v .
- If K does not halt on x , then $\text{nsteps}(M, w) = \infty = \text{nsteps}(N, v)$; so $\text{nsteps}(M, w) \geq \text{nsteps}(N, v)$, for all w, v .

So, L_c is non recursive.

(d) non-R.E.

To show $\overline{\text{HP}} \leq_m L_d$:

$$K \# x \mapsto M \# N$$

$$K \text{ does not halt on } x \iff \text{nsteps}(M, w) < \text{nsteps}(N, v) \quad \forall w, v$$

$$K \text{ does not halt on } x \Rightarrow \forall w, v \quad \text{nsteps}(M, w) < \text{nsteps}(N, v)$$

$$K \text{ halts on } x \Rightarrow \exists w, v \quad \text{nsteps}(M, w) \geq \text{nsteps}(N, v)$$

M on input w :

- Simulate K on x for $|w|$ steps.
- If K did not halt yet, accept w (and halt). Otherwise, loop.

N on input v :

- Simulate K on x .
- If it halts, accept v .

If K does not halt on x , then

- M will halt on all strings in finite steps.
- N will loop on every string.
- So, we have $\forall w, v \quad \text{nsteps}(M, w) < \text{nsteps}(N, v)$

If K halts on x , say in s steps:

- M loops on strings w , with $|w| \geq s$.
- N will halts on all strings in finite steps.
- So, we have $\exists w, v \quad \text{nsteps}(M, w) \geq \text{nsteps}(N, v)$

Thus, the reduction is correct.

Therefore, L_d is non-R.E. □

Problem 6 Prove that the following languages are not recursive.

- (a) $L_1 = \{M \# N \mid L(M) = L(N)\}$
- (b) $L_2 = \{M \# N \mid L(M) \subseteq L(N)\}$
- (c) $L_3 = \{M \# N \mid L(M) \cap L(N) = \phi\}$
- (d) $L_4 = \{M \# N \# P \mid L(M) \cap L(N) = L(P)\}$

Solution (a) To show $\text{HP} \leq_m L_1$:

$$K \# w \mapsto M \# N$$

$$K \text{ halts on } w \iff L(M) = L(N)$$

M on input v_1 :

- Accept v_1 .

N on input v_2 :

- Simulate K on w .
- If K halts, then accept v_2 .

$$L(M) = \Sigma^*$$

$$L(N) = \begin{cases} \Sigma^* & \text{if } K \text{ halts on } w \\ \phi & \text{if } K \text{ does not halt on } w \end{cases}$$

So, if K halts on w , $L(M) = L(N)$ and if K does not halt on w , then $L(M) \neq L(N)$, completing the reduction. Thus, L_1 is non-recursive.

(b) To show $\text{HP} \leq_m L_2$.

Use the same construction as in (a).

If K halts on w , $L(M) \subseteq L(N)$, and if K does not halt on w , then $L(M) \not\subseteq L(N)$, completing the reduction. Thus, L_2 is non-recursive.

(c) Here we will show $\overline{\text{HP}} \leq_m L_3$.

$$K \# w \mapsto M \# N$$

$$K \text{ does not halt on } w \iff L(M) \cap L(N) = \phi$$

$$K \text{ does not halt on } w \Rightarrow L(M) \cap L(N) = \phi$$

$$K \text{ halts on } w \Rightarrow L(M) \cap L(N) \neq \phi$$

Again use the same construction as in (a).

If K halts on w , $L(M) \cap L(N) = \Sigma^* \neq \phi$. If K does not halt on w , then $L(M) \cap L(N) = \phi$.

Thus, L_3 is non-R.E., and consequently also non-recursive.

(d) To show $\overline{\text{HP}} \leq_m L_4$.

$$K \# w \mapsto M \# N \# P$$

$$K \text{ does not halt on } w \iff L(M) \cap L(N) = L(P)$$

$$K \text{ does not halt on } w \Rightarrow L(M) \cap L(N) = L(P)$$

$$K \text{ halts on } w \Rightarrow L(M) \cap L(N) \neq L(P)$$

M on input v_1 :

- Simulate K on w .
- If K halts, accept v_1 .

N on input v_2 :

- Accept v_2

P on input v_3 :

- Reject v_3 .

Here,

$$L(M) = \begin{cases} \Sigma^* & \text{if } K \text{ halts on } w \\ \phi & \text{if } K \text{ does not halt on } w \end{cases}$$

$$L(N) = \Sigma^*$$

$$L(P) = \phi$$

So, if K halts on w , $L(M) \cap L(N) = \Sigma^* \neq \phi = L(P)$. If K does not halt on w , then $L(M) \cap L(N) = \phi = L(P)$.

Thus, L_4 is non-R.E. and thus consequently non-recursive. □

Problem 7 Prove that the following problems on a TM M are decidable.

- (a) Decide whether M halts on some input within 2026 steps.
- (b) Decide whether M on a given input w moves left at least 2026 times.

Solution (a) Simulate M on all possible inputs w , $|w| \leq 2026$, parallely, for 2026 steps. If it halts on any one of them (within 2026 steps), then accept M . Otherwise, reject.

Why does this work?

- **Case 1:** M does not halt on any input within 2026 steps. Then our decider correctly rejects M (since it won't halt in 2026 steps on our chosen inputs).
 - **Case 2:** M halts on some input w^* within 2026 steps.
 - If $|w^*| \leq 2026$, then again our decider correctly accepts M since its simulation of w^* will halt within 2026 steps.
 - If $|w^*| > 2026$, and we know that M halts on w^* within 2026 steps. This must mean that M only ever sees at most 2026 characters of w^* from the start (to see more characters, it would require more steps). Then, consider the prefix α of w^* consisting of its first 2026 characters. The behaviour of M on α should be identical to that on w^* , since the part of the input tape which is reachable by the machine is identical. Therefore, M halts on α within 2026 steps. Hence, our decider correctly accepts M , since its simulation of α will halt within 2026 steps.
- (b) Simulate M on w for $2026 \times (|w| + |Q|)$ steps. During the simulation, if M moves left at least 2026 times, accept. Otherwise, reject.

Why does this work? Between any two L moves that M makes, what is the maximum number of R moves it can take? If it is at the start of the tape (at the $\vdash$), it can go right for $|w|$ steps, reaching the end of the string. Then, it can go further right, finding a blank symbol, say at state q_1 . It may move right again, going to some state q_2 . If it keeps moving right, at some point it will repeat states and be stuck in a loop, forever moving right (because it keeps seeing $\square$ symbols). So, we can bound the number of consecutive right moves M can take before taking a left move, to be $|w| + |Q|$. So between two L moves, there can be at most $|w| + |Q|$ R moves. Thus, it follows that if M has at least k left moves, they must happen within $k(|w| + |Q|)$ steps. $\square$

Problem 8 Use Rice's theorems to prove that neither the following languages nor their complements are R.E.

- (a) $\text{FIN} = \{M \mid L(M) \text{ is finite}\}$.
- (b) $\text{REG} = \{M \mid L(M) \text{ is regular}\}$.

Solution (a) FIN is non monotone, since $\phi \subseteq \Sigma^*$, but $P_{\text{FIN}}(\phi) = T$, and $P_{\text{FIN}}(\Sigma^*) = F$. Therefore by Rice's theorem, FIN is non-R.E.

$\overline{\text{FIN}}$ is monotone, so we cannot comment on it by Rice theorem. We must show $\overline{\text{HP}} \leq_m \overline{\text{FIN}}$ to prove it is non-R.E.

- (b) Let $A = \{a^p \mid p \text{ is prime}\}$.

REG is non monotone, since $\phi \subseteq A$, but $P_{\text{REG}}(\phi) = T$ and $P_{\text{REG}}(A) = F$. Therefore by Rice's theorem, REG is non-R.E.

$\overline{\text{REG}}$ is non monotone, since $A \subseteq \Sigma^*$, but $P_{\overline{\text{REG}}}(A) = T$ and $P_{\overline{\text{REG}}}(\Sigma^*) = F$. Therefore by Rice's theorem, $\overline{\text{REG}}$ is non-R.E. $\square$

Problem 9 Prove/disprove: No non-trivial property of R.E. languages is semidecidable.

Solution Counter example: Non-emptiness.

$\overline{(E_{\text{TM}})} = \{M \mid L(M) \neq \phi\}$ is a non trivial property, which is semidecidable.) $\square$
